

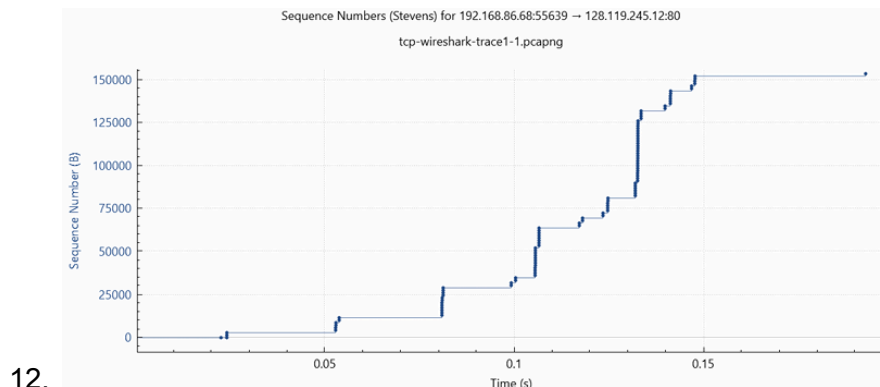
UDP

1. 5, carrying an **SSDP (Simple Service Discovery Protocol)** message. 4 fields in UDP header: source port, destination port, length, check sum
2. 2,2,2,2 bytes (8 bytes total)
3. Value of entire length field is the value of entire UDP segment. Which is 283. Payload = 275, header = 8
4. $2^{16}-1 = 65535$ (16 bits length field)
5. Largest possible source port = 65535
6. UDP Protocol number: 17
7. For packet 15
Src: 58350
Dest: 53
For packet 17
Src: 53
Dest:
58350

TCP

1. Client IP Address (Source): 192.168.86.68
Client TCP Port Number (Source): 55639
2. 128.119.245.12, port = 80
3. Sequence number (raw): 4236649187, The **Flags** field is shown as 0x002 (SYN), yes (because in options, there are a sack permission)
4. Sequence number (raw): 1068969752, flag: 0x012 (SYN, ACK), acknowledgement number (raw) : 4236649188, by adding one so (87 -> 88)
5. Sequence number (raw): 4236801228, 1385 bytes (tcp payload), no, we need multiple segments

6. At time = 0.024047 the first segment in the data-transfer part of the TCP connection was sent. At time=0.052671 the ACK for this first data-containing segment was received
 $RTT_1 = 0.052671 - 0.024047 = 0.028624$
 $RTT_2 = 0.052676 - 0.024048 = 0.028628$
7. the length (header plus payload) of each of the first four data-carrying TCP segments is 1514.
8. the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments is 31872 bytes.
9. No, there are no retransmitted segments in the trace file. i checked the whole file of data packets to find this. Data packet with seq no 17377 was lost as the ack for this packet never arrived, but this was not retransmitted in the file.
10. 1448 bytes of data the receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to server.
11. $0.192625 - 0.024047 = 0.168578$
Throughput: $153,425 / 0.16857 = 909,641$ bytes/s



13. differences between the fleet times listed:

- $0.053 - 0.025 \approx 0.028$ s
- $0.082 - 0.053 \approx 0.029$ s
- $0.100 - 0.082 \approx 0.018$ s (one is a bit shorter, possibly measurement/granularity or slightly earlier retransmit/packing)

The two largest gaps give an average of about ~ 0.028 – 0.03 seconds (≈ 28 – 30 ms). This period is the round-trip time (RTT) for this connection — each fleet is spaced by roughly one RTT